

ResumeOS Constraint-Aware Resume Intelligence Platform Master PRD + Technical Architecture + Workflow Documentation

This document combines the Product Requirements Document (PRD), technical architecture, workflow systems, rendering pipeline design, AI orchestration strategy, ATS optimization framework, and engineering roadmap for ResumeOS — an AI-powered constraint-aware resume intelligence platform. The uploaded resume template demonstrates a standardized MBA-style placement resume with tightly controlled sections, typography, hierarchy, and whitespace management. The platform architecture is optimized around preserving these formatting constraints while enabling AI-powered semantic optimization.

1. Product Vision

ResumeOS is not a conventional AI resume builder. It is a structured document intelligence system capable of: - understanding document layouts, - preserving typography and spacing, - optimizing resume content semantically, - validating ATS compatibility, - and generating recruiter-aware alternatives without breaking formatting. The uploaded sample resume confirms the need for strict structure preservation: - one-page MBA layout, - fixed hierarchy, - standardized bullet structure, - institutional formatting consistency, - tightly controlled whitespace density.

2. Core Product Thesis

Most AI resume systems optimize semantics but destroy formatting. ResumeOS introduces: AI optimization under strict visual constraints. Every uploaded resume becomes a: Document DNA Schema. This schema captures: - typography - margins - line spacing - section widths - indentation - bullet alignment - section ordering - visual rhythm - whitespace density All future AI modifications must obey these constraints.

3. Resume Template Intelligence

The uploaded resume sample reveals recurring structural patterns likely common across institutional resumes: Detected Characteristics: - single-page compact structure - highly compressed vertical spacing - recruiter-oriented density - institutional section ordering - bullet-driven achievement formatting - quantified achievements - multi-level section hierarchy Expected Generalization: Future resumes may contain: - additional sections, - fewer sections, - reordered sections, - altered content density, but will likely preserve: - typography families, - structural hierarchy, -

formatting rhythm, - alignment behavior. This allows ResumeOS to use: Template Family Recognition.

4. Product Modules

A. Document DNA Engine Extracts formatting metadata and layout constraints. B. Layout Constraint Engine Prevents formatting drift after AI modifications. C. Gemini Rewrite Engine Generates ATS-aware semantic alternatives. D. ATS Consensus Engine Validates compatibility across ATS systems. E. Recruiter Simulation Engine Simulates recruiter scanning behavior. F. Resume Diff Engine Visualizes semantic and formatting changes. G. Live Rendering Engine Provides near-real-time layout previews. H. Persona Optimization Engine Generates role-specific resume variants.

5. Technical Architecture

Recommended Stack: Frontend: - Next.js - React - Tailwind CSS - Canvas rendering layer Backend: - Python FastAPI Database: - Supabase PostgreSQL Storage: - Supabase Storage Authentication: - Supabase Auth AI: - Gemini 2.5 Flash - Gemini 2.5 Pro Rendering: - LibreOffice Headless - python-docx - lxml - fonttools Deployment: - Vercel (Frontend) - Railway or Render (Backend)

6. Why This Architecture

The system is fundamentally: document-processing heavy. Therefore: Python is essential. Reasons: - DOCX manipulation ecosystem - PDF generation - typography analysis - rendering pipelines - XML parsing - AI orchestration workflows Supabase is preferred over Firebase because: - structured relational data, - SQL querying, - pgvector support, - lower entry restrictions, - better suitability for resume schemas.

7. Workflow Architecture

Workflow Pipeline: Step 1: User uploads resume. Step 2: Document DNA Engine extracts: - sections - typography - margins - bullets - indentation - spacing rules Step 3: System generates structured schema. Step 4: User uploads JD. Step 5: Gemini extracts: - keywords - skills - competencies - recruiter intent Step 6: Constraint-aware AI generation begins. Step 7: Generated bullets validated against: - width constraints - line wrapping - layout preservation Step 8: ATS validation engine runs. Step 9: User interactively selects alternatives. Step 10: System exports DOCX and PDF.

8. Document DNA Schema

Each uploaded resume generates: Template Metadata: - page dimensions - section map - paragraph styles
Typography Metadata: - font family - font size - weight - casing Spatial Metadata: - line width - wrapping thresholds -
whitespace zones Semantic Metadata: - achievements - metrics - leadership signals - action verbs Constraint
Metadata: - max lines per bullet - section height ceilings - spacing preservation rules

9. Constraint-Aware Generation

This is the platform's core innovation. The system must generate: - ATS optimized, - recruiter optimized, -
semantically accurate text while preserving: - identical line count, - identical section height, - identical layout behavior.
Important Insight: Character count does not equal visual width. The engine must therefore: measure rendered width
using typography metrics.

10. Rendering & Validation Engine

Rendering is the hardest engineering problem. The system must detect: - line overflow - wrapping changes - page
shifts - spacing distortion Recommended Strategy: Approximate rendering in browser. Exact rendering validation on
backend. Initial Rendering Engine: LibreOffice Headless. This avoids building a full Word rendering engine from
scratch.

11. ATS Validation Layer

ATS integrations: - Affinda - RChilli - Sovren - Textkernel Validation Checks: - section parsing - experience extraction -
skill recognition - formatting compatibility - keyword extraction Future Feature: Multi-ATS consensus scoring.

12. Recruiter Intelligence Layer

The platform should simulate: - consulting recruiter - startup recruiter - PM recruiter - HR recruiter - analytics recruiter
Metrics: - readability - leadership density - quantified impact - scan efficiency - semantic strength - clutter score

13. Creative Product Features

Resume DNA Lock: Prevents formatting drift. Semantic Density Meter: Measures informational value per line.
Recruiter Heatmap: Predicts recruiter attention zones. Resume Compression Intelligence: Compresses verbose
content while preserving meaning. Anti-AI Layer: Reduces generic AI phrasing. Persona Switching: Generate

consulting/PM/analytics variants. Resume Pressure Testing: Simulate recruiter and ATS outcomes.

14. Suggested Database Schema

Core Tables: users resumes resume_templates resume_sections resume_bullets job_descriptions ats_reports semantic_scores layout_constraints resume_versions Important Principle: Store resumes as structured systems, not just uploaded files.

15. API Architecture

Core APIs: POST /upload-resume POST /extract-schema POST /analyze-jd POST /generate-alternatives POST /validate-layout POST /ats-check POST /export-docx POST /export-pdf Future APIs: POST /recruiter-simulation POST /persona-switch POST /semantic-density-analysis

16. Claude Code Engineering Roadmap

Priority Order: 1. Resume parser 2. Schema extraction 3. Typography engine 4. Layout validator 5. Gemini rewriting engine 6. Live preview renderer 7. ATS integrations 8. Recruiter simulation Important Engineering Warning: Do NOT prioritize UI before rendering fidelity is solved.

17. Security & Privacy

Resumes contain sensitive information. Requirements: - encrypted storage - secure file handling - signed URLs - deletion workflows - role-based access Future enterprise deployments may require: - GDPR alignment - SOC2 readiness - institutional compliance.

18. MVP Scope

MVP Includes: - DOCX/PDF upload - schema extraction - JD analysis - Gemini optimization - overflow validation - DOCX export - PDF export MVP Excludes: - multiplayer editing - mobile apps - Chrome extension - enterprise ATS syncing - advanced analytics

19. Future Vision

ResumeOS evolves into: AI Operating System for Professional Documents. Future supported assets: - consulting slides - SOPs - LinkedIn optimization - academic CVs - government forms - procurement documents - grant applications The deeper moat is: document layout intelligence.

20. Final Positioning

Do NOT position ResumeOS as: “AI Resume Builder.” Position it as: - Constraint-Aware Resume Intelligence Platform - Structured Career Document Intelligence System - AI Resume Optimization Infrastructure The category itself should feel new.
